

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: B | Time: 3 Hours | Max Marks: 40 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which SQL keyword is used to retrieve data from a database? [2 Marks]
 - a) UPDATE
 - b) INSERT
 - c) SELECT
 - d) DELETE

2. A database constraint that ensures that a value in one table's column(s) must match a value in another (parent) table's primary key column(s) is known as a: [2 Marks]
 - a) Primary key constraint
 - b) Unique constraint
 - c) Foreign key constraint
 - d) Not Null constraint

3. Which of the following is an example of a DDL (Data Definition Language) command? [2 Marks]
 - a) INSERT
 - b) UPDATE
 - c) DELETE
 - d) ALTER

4. The phenomenon where a transaction reads uncommitted data written by another transaction, which then subsequently rolls back, is known as: [2 Marks]
 - a) Lost Update
 - b) Dirty Read
 - c) Non-repeatable Read
 - d) Phantom Read

5. In the context of a relational database, what does a "tuple" refer to? [2 Marks]
 - a) A column in a table
 - b) A row in a table
 - c) A database schema
 - d) An index

6. Which of the following database components is responsible for managing the allocation of space on disk storage and the data structures used for data storage? [2 Marks]

- a) Query Processor
- b) Transaction Manager
- c) Storage Manager
- d) Buffer Manager

SECTION A: SHORT ANSWERS

7. Explain the role of a Database Administrator (DBA) in managing a DBMS. List two key responsibilities. **[6 Marks]**
8. Differentiate between a DDL (Data Definition Language) and a DML (Data Manipulation Language) in a DBMS, providing an example command for each. **[5 Marks]**
9. What is a database schema? Briefly explain the three-level architecture (ANSI/SPARC architecture) for a DBMS. **[4 Marks]**
10. Describe the "Lost Update" problem that can occur in a multi-user database environment without proper concurrency control. Provide a simple example scenario. **[7 Marks]**
11. Briefly explain the purpose of database normalization. Name and briefly describe the first normal form (1NF). **[6 Marks]**

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